

Sathvik S Rao

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PROFESSIONAL SUMMARY

Final-year Computer Science undergraduate (CGPA: 9.25) with hands-on experience building full-stack and applied machine learning systems, including computer vision pipelines and cloud-deployed backends. Published IEEE researcher and active technical community contributor, seeking software engineering / ML engineering roles.

EDUCATION

SRM Institute of Technology

Bachelor of Engineering in Computer Science and Engineering; CGPA: 9.25

- Relevant Coursework: Data Structures and Algorithms, Internet of Things

Tamil Nadu, India

Jun 2023 – May 2027 (Expected)

SELECT PROJECTS

PokePrice | *React Native (Expo), Python, FastAPI, PyTorch, OpenCV, EasyOCR, Redux Toolkit*

- Architected an end-to-end Pokémon card identification platform: a React Native (Expo) mobile client capturing card photos, paired with a FastAPI backend that performs OpenCV-based perspective correction, ConvNeXt V2 classification, and EasyOCR text extraction to identify 100+ unique card classes
- Trained a ConvNeXt V2 image classifier in PyTorch on 20,000+ labeled Pokémon card images, applying transfer learning and data augmentation to reach ~90% recognition accuracy
- Tuned EasyOCR with character allowlisting to reduce misreads on stylized card fonts and speed up CPU-only text extraction used for card name/number verification
- Optimized on-device image preprocessing with Expo Image Manipulator (resolution capping and compression), cutting image payload size by ~60% and improving end-to-end API response time by ~40%
- Designed a RESTful API connecting the mobile frontend to the ML backend, delivering card identification and live pricing results in under 2 seconds on average
- Implemented Redux Toolkit and AsyncStorage to cache scan history locally, cutting redundant API calls by ~50% and enabling offline access to past scans

Big-Tac-Toe | *HTML/CSS/JavaScript, Java Spring Boot, MongoDB Atlas | Azure*

- Built a single-player mode against an 'AI opponent' using Monte Carlo Tree Search (MCTS) for strategic move selection
- Developed a responsive, cross-browser frontend with HTML5, CSS3 (Flexbox/Grid), and ES6+ JavaScript, optimized for both mobile and desktop
- Built and deployed RESTful backend APIs with Java Spring Boot on Microsoft Azure to manage multiplayer sessions and game logic
- Integrated MongoDB Atlas to persist user data and game history with high availability and scalability

PUBLICATIONS

Sathvik S Rao, Shayantan Bose, Ansh Baiju, Dr. Shajeena Elmo, “A Comparative Analysis on the Differences Between Client-Side Databases and Server-Side Databases,” IEEE (2025)

TECHNICAL SKILLS

Languages: Java, C++, Python, JavaScript, SQL, HTML/CSS

Frameworks & Libraries: Spring Boot, FastAPI, PyTorch, OpenCV, React Native (Expo), Redux Toolkit

Tools & Platforms: Git/GitHub, MongoDB Atlas, Microsoft Azure

LEADERSHIP / EXTRACURRICULAR

Robothinkers Club – Member: co-organized technical events and delivered a seminar on Arduino fundamentals to fellow students